Digital Twin Framework for Industrial Assets Prognostics

Using NASA Turbofan Engine Dataset

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Masters Capstone Project

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1 Problem Statement & Motivation

Predicting Failures Before They Happen, Not After.

Industrial assets such as aircraft engines operate under complex conditions and degrade over time. Unexpected failures lead to safety risks, high maintenance costs, and unplanned downtime.

Key Challenges

Unexpected Failures
Hard to predict engine failures in advance.

High Maintenance Cost
Reactive maintenance leads to expensive repairs.

Downtime & Disruptions
Unplanned downtime affects operations and reliability.

Limited Visibility
Difficult to monitor internal degradation processes.

Goal

Develop a data-driven framework to accurately predict the **Remaining Useful Life (RUL)** of turbofan engines using the NASA Turbofan Engine Degradation Simulation Dataset.

Why It Matters

Safety
Prevent catastrophic failures and ensure operational safety.

Cost Efficiency
Reduce maintenance costs through predictive insights.

Operational Reliability
Improve asset availability and mission reliability.

Sustainability
Optimize resource use and reduce waste.

Our Approach

Data-Driven Digital Twin → AI/ML Prognostics → Proactive Maintenance → Reduced Cost & Downtime

Motivation

By leveraging digital twin technology and machine learning, we can shift from **reactive** maintenance, extend asset life, and ensure **safer, more efficient** operations.

Predict Early | Optimize Maintenance | Reduce Downtime | Lower Costs

2 Digital Twin Concept

A Virtual Representation. Real-time Data. Smarter Decisions.

A Digital Twin is a virtual representation of a physical asset that uses real-time data to simulate, monitor, and predict the behavior and health of the system.

How It Works

1. Physical Engine
Real turbofan engine operating in real-world conditions.

2. Sensor Data
Sensors collect data on various parameters (temperature, pressure, speed, vibration, etc.).

3. Data Ingestion
Data is transmitted and stored for processing and analysis.

4. Digital Twin (Model)
ML models learn system behavior and simulate degradation over time.

5. Prognostics Output
Predict Remaining Useful Life (RUL) and health of the engine.

In Our Framework

NASA Turbofan Engine Dataset → ML/DL Models → Health Estimation → RUL Prediction → Proactive Maintenance

Benefits of Digital Twin

Predict Failures Early
Identify potential failures before they occur.

Optimize Maintenance
Enable condition-based maintenance and reduce unnecessary servicing.

Reduce Costs & Downtime
Minimize unexpected breakdowns, repair costs, and operational downtime.

Improve Safety & Reliability
Enhance system reliability and ensure operational safety.

Data-Driven Decisions
Support smarter and faster decision making with real-time insights.

Example: Turbofan Engine

- Temperature
- Pressure
- Speed (Nf, Nc)
- Exhaust Gas Temp (EGT)
- Vibration
- Fuel Flow
- Operational Settings
- ... and more

All these parameters are used by the digital twin to simulate engine behavior and predict **Remaining Useful Life (RUL).**

Key Takeaway

A Digital Twin acts as a bridge between the physical world and intelligent decisions, enabling proactive maintenance, cost savings, and improved operational efficiency.

3 Dataset Overview

We use the NASA Turbofan Engine Degradation Simulation Dataset – **FD001** for developing and evaluating our prognostics framework.

Source
NASA Turbofan Engine Degradation Simulation Data (FD001)

Engine
Simulated turbofan engines under various operating conditions

Target
Remaining Useful Life (RUL)

Unit of Analysis
Engine run-to-failure (cycle-by-cycle)

Sensors
21 sensors + 3 operational settings

Dataset Snapshot (FD001)

Train Rows	20,361
Test Rows	13,096
Train Engines	100
Test Engines	100
Operating Conditions	1

Multivariate time-series data capturing engine degradation patterns

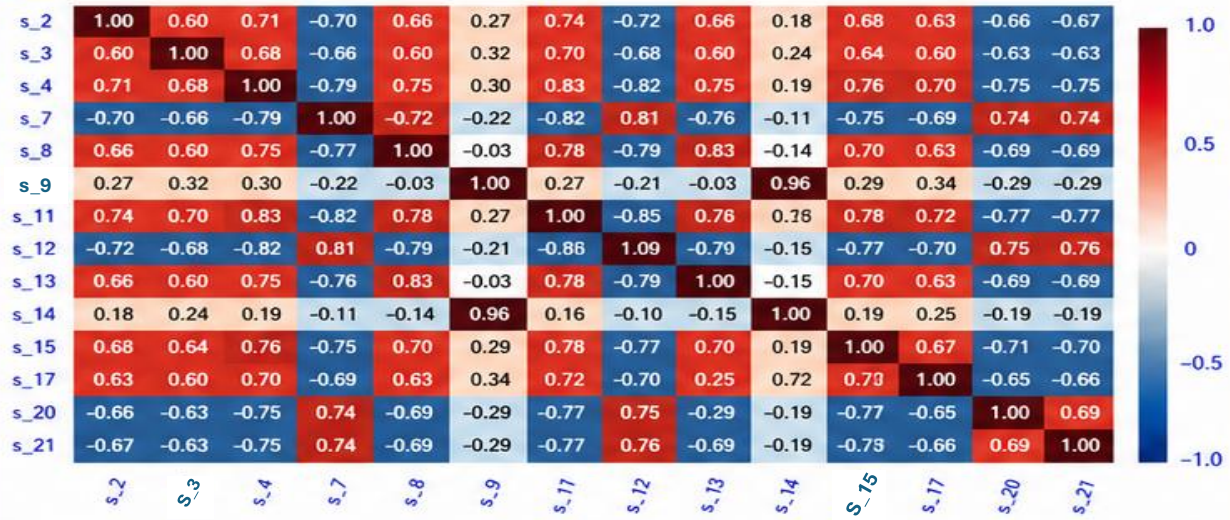
Key Insights

- FD001 provides run-to-failure data from a single operating condition.
- It contains multiple engines with varying degradation patterns.
- The dataset is widely used for Remaining Useful Life (RUL) prediction.

4 Exploratory Data Analysis (EDA)

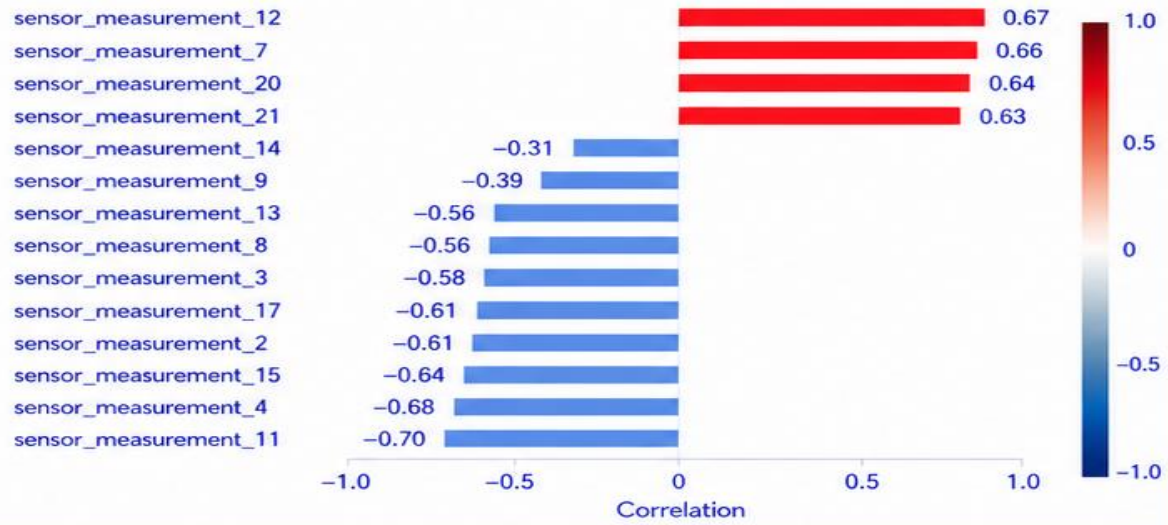
Understanding the data, uncovering patterns & relationships.

Sensor Correlation Heatmap



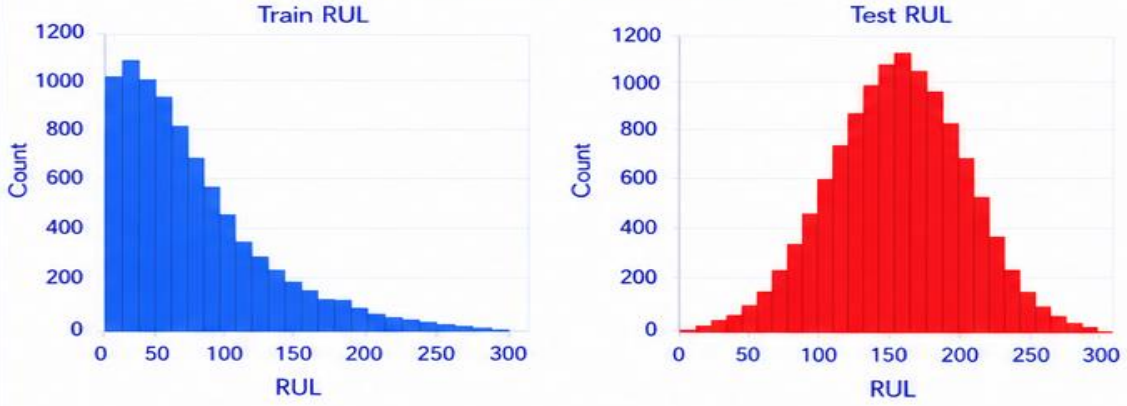
Sensors show clustered dependencies and redundancy, strong coupled degradation behavior.

Correlation of Sensors with RUL (Ranked)



A subset of sensors has strong positive or negative association with RUL, highlighting informative degradation signals.

RUL Distribution: Train vs Test



Key Insights

RUL Distribution (Train vs Test)

RUL is non-uniform (right-skewed with denser low-to-mid regions), so prediction difficulty is regime-dependent; this supports capped targets and uncertainty-aware reporting.

Sensor Correlation Heatmap

Sensors show clustered dependencies and redundancy, indicating strong coupled degradation behavior; this justifies feature cleanup and engineering rather than treating all raw channels equally.

Sensor Correlation with RUL (Ranked)

A subset of sensors has materially stronger association with RUL (both positive and negative trends), showing that informative degradation signals exist and motivating for_*, rc_*, rs_*_*, si_*) and explainability.

5 Data Preprocessing & Feature Engineering

Transforming raw engine signals into reliable sequence inputs for RUL learning

Preprocessing Steps

Removed low-information channels
Excluded constant/weakly informative operating and sensor signals.

Built capped RUL target
Focused learning on decision-relevant degradation range.

Applied per-engine smoothing
Reduced sensor noise while preserving degradation trends.

Standardized features consistently
Scaled features to stable ranges for model training.

Feature Engineering

Trend features (tr_*)
Captured slope-based degradation direction.

Rate-of-change features (rc_*)
Measured short-term dynamic shifts.

Rolling-stat features (rs_*)
Summarized local behavior over recent cycles.

Interaction features (si_*)
Modeled coupled effects between sensors.

Modeling-Ready Result

Sequence representation created
Sliding windows preserve temporal context.

Temporal dependencies retained
Inputs reflect evolution, not isolated snapshots.

Noise-robust, informative feature space
Cleaner signal for stable learning.

Prepared for quantile RUL prediction
Enables low/median/high life estimation with uncertainty.

Key Takeaway: Careful preprocessing and temporal feature engineering convert noisy raw sensor streams into structured, informative sequences that improve RUL learning quality and uncertainty-aware prediction reliability.

Clean Data | Engineered Features | Better Learning | Uncertainty-Aware Prediction

6 Modeling Approach & Inference Pipeline

End-to-end quantile LSTM pipeline for accurate, uncertainty-aware RUL prediction on NASA C-MAPSS FD001

1. FD001 multivariate engine cycles
Raw sensor readings over time

→

2. Preprocessing
Cleanup, RUL cap, per-engine smoothing, scaling

→

3. Feature engineering
Create tr_*, rc_*, rs_*, si_* features

→

4. Sequence windowing
seq_len = 30; sequence tensor

→

5. Quantile LSTM
Model predicts quantiles: q0.1, q0.5, q0.9

→

6. Uncertainty derivation
Derive low / mid / high RUL and interval width

→

7. Confidence proxy
confidence = exp(-width/k); k from median width

→

8. Explainability
SHAP values for median head

→

9. Engine-level evaluation
Evaluate using last-window-per-test-engine

A. Quantile LSTM Configuration

- Sequence input and temporal modeling
- Quantile outputs: rul_low, rul_mid, rul_high
- Pinball (quantile) loss
- Decision-oriented interval prediction

B. Evaluation Metrics

- RMSE
- R^2
- NASA score
- weighted MAE
- coverage
- mean interval width
- within_10_pct
- within_20_pct

C. Operational Decision Logic

- width = rul_high - rul_low
- confidence = exp(-width/k)
- confidence is ranking proxy, not calibrated probability
- supports urgency + uncertainty triage

Key Takeaway: Single quantile LSTM pipeline integrates accuracy, uncertainty, and explainability into decision-ready prognostics.

7 Final Model Architecture (Quantile LSTM)

Uncertainty-aware RUL prediction with quantile outputs on NASA C-MAPSS FD001

Model Architecture

Input Sequence Window
(seq_len = 30, n_features = 66)
multivariate engineered features

↓

LSTM Layer 1
LSTM(32, return_sequences=True)
captures short/medium temporal patterns

↓

LSTM Layer 2
LSTM(16)
refines temporal representation

Dense Layer
Dense(16, activation='relu')
learns compact representation

Dropout Layer
Dropout(0.2)
reduces overfitting

Quantile Output Layer
Dense(3, activation='linear')
outputs [q0.1, q0.5, q0.9]

↓

Prediction Outputs (3 Quantiles)

rul_low (q0.1) | rul_mid (q0.5) | rul_high (q0.9)

Key Components & Design Choices

Sequence LSTM layers
model temporal degradation dynamics.

Quantile head
predicts lower/median/upper RUL simultaneously (q0.1, q0.5, q0.9).

Training objective: quantile (pinball) loss
optimizes interval accuracy across quantiles.

Uncertainty interval
derived from rul_high - rul_low.

Confidence derived from interval width
confidence = exp(-width/k)
(ranking proxy, not probability).

SHAP interpretation
is mainly on the median head (q0.5) to explain predictions.

Training Configuration

- **Loss:** quantile (pinball) loss over [0.1, 0.5, 0.9]
- **Optimizer:** Adam

Key Takeaway: Single quantile LSTM pipeline integrates accuracy, uncertainty, and explainability into decision-ready prognostics.

8 Final Model Performance

Uncertainty-aware quantile LSTM performance on NASA C-MAPSS FD001

Performance on Test Set (last-window-per-test-engine)

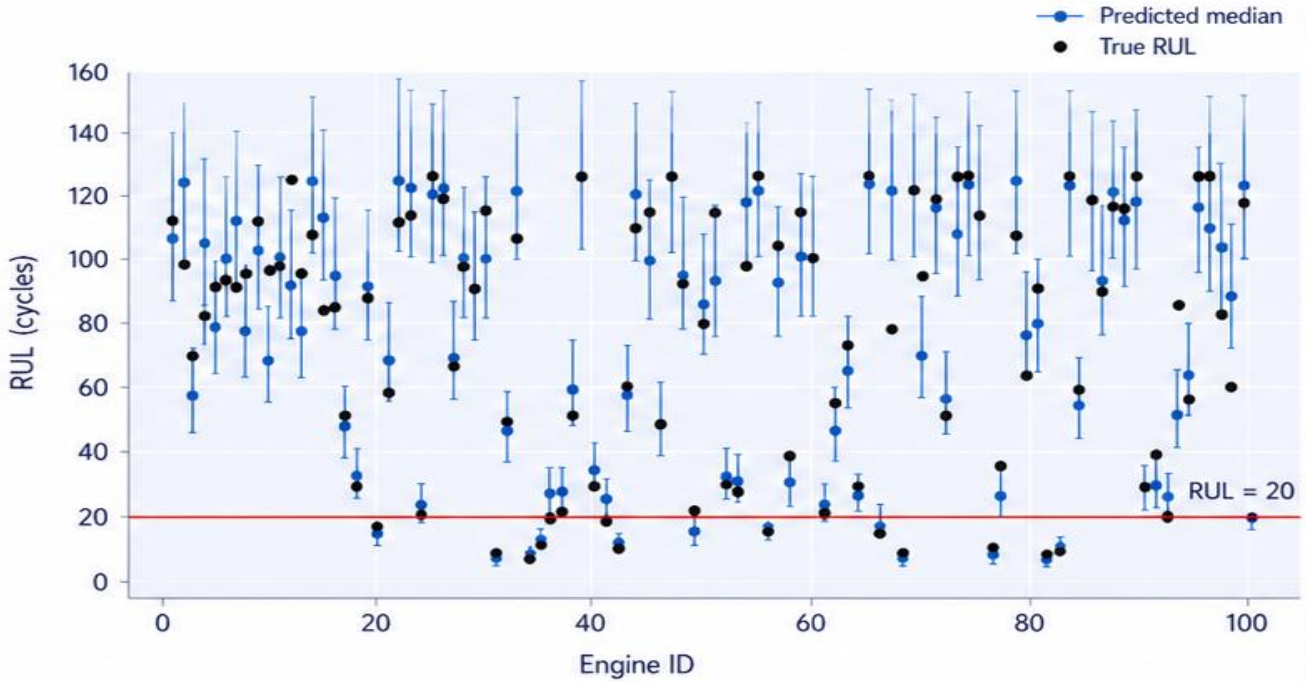
Metric	Value
RMSE (cycles)	12.51
R^2 Score	0.903
NASA Score (↓)	274.41
Weighted MAE (cycles)	7.22
Coverage (within interval)	0.83
Mean Interval Width (cycles)	32.86

Phase-aware Performance (RMSE / Within10%)

Phase	Scope	RMSE (cycles)	Within10%
≤20 Critical Phase	RUL ≤ 20 (n = 16)	3.0735	100.00%
>20 Healthy Phase	RUL > 20 (n = 84)	13.5835	61.90%

Scope: last-window-per-test-engine (n = 100)

Per-engine Predicted Interval (Low–High) vs True RUL



Strong Central Prediction Fidelity

High correlation with true RUL achieved (R^2 = 0.903) and low weighted error (wMAE = 7.22 cycles).

Useful Uncertainty Quantification

Good interval coverage (0.83) with interpretable mean interval width (32.86 cycles) across engines.

Decision Utility via Confidence

Confidence derived from interval width enables risk-aware ranking and effective triage of engines.

Bottom Line: The quantile LSTM delivers accurate and uncertainty-aware RUL predictions, with perfect critical-phase detection (100% within 10% for RUL ≤ 20) and acceptable performance in the healthy phase.

Note: Lower is better for RMSE, NASA Score, Weighted MAE, and Mean Interval Width. Higher is better for R^2, Coverage, Within10%.

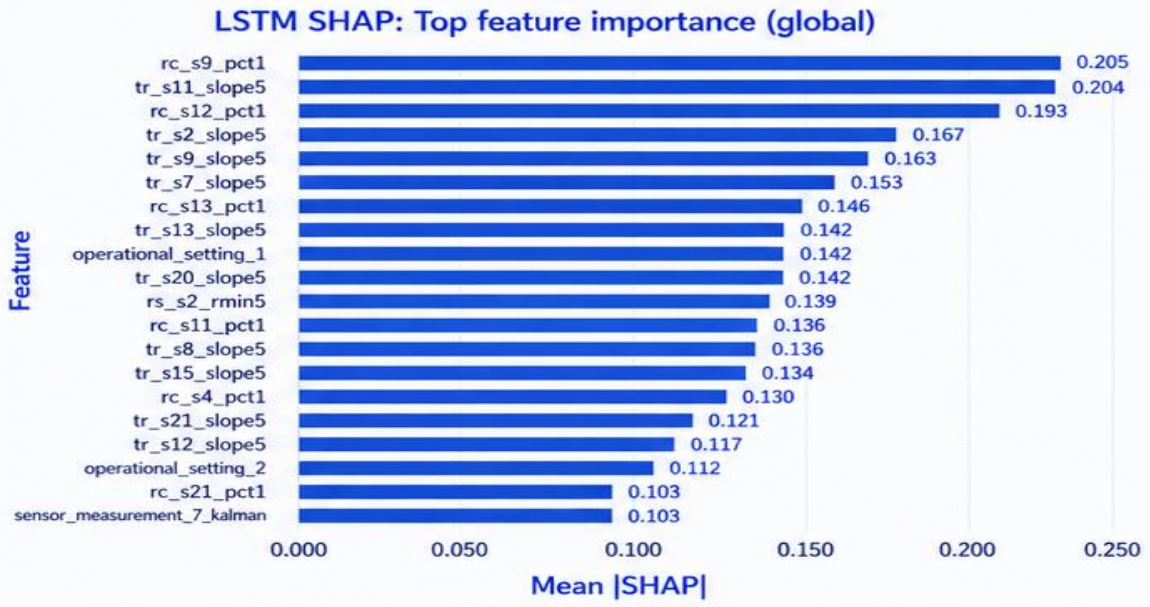
9 Key Insights & Findings

Quantile LSTM achieved strong point performance on FD001 (RMSE **12.51**, R^2 **0.903**, NASA **274.41**) under last-window-per-engine evaluation.

Interval predictions added decision context (coverage **0.83**, mean interval width **32.86**), enabling uncertainty-aware triage.

Performance is stronger near critical RUL and less sharp in healthier high-RUL regimes (expected PHM behavior).

SHAP (median head) showed trend/rate-of-change dominated features, supporting model interpretability.



Literature Comparison (FD001, Last-Window-Per-Engine Evaluation)

Model	NASA Score (↓)	RMSE (cycles) (↓)	Notes
Attention-LSTM [1]	200.00	12.33	Uses attention mechanism with LSTM
Proposed Quantile LSTM (Mine)	**274.41**	**12.51**	**Uncertainty-aware with quantile outputs (q0.1, q0.5, q0.9)**
CAELSTM [2]	282.38	14.44	Convolutional autoencoder + LSTM
Stacked LSTM [3]	311.20	15.22	Deep stacked LSTM architecture

Lower is better for NASA Score and RMSE. Results may vary slightly due to different preprecessing and evaluation settings.

[1] M. Dida, A. Cheriet, and M. Belhadj, "Remaining Useful Life Prediction Using Attention-LSTM Neural Network of Aircraft Engines," Int. J. Progn. Health Manag., vol. 16, no. 2, 2025. [Online]. Available: https://papers.phmsociety.org/index.php/ijphm/article/view/4274

[2] S. M. Elsherif, B. Hafiz, M. A. Makhlouf, et al., "A deep learning-based prognostic approach for predicting turbofan engine degradation and remaining useful life," Sci. Rep., vol. 15, Art. no. 26251, 2025, doi: 10.1038/s41598-025-09155-z.

[3] T. S. Syuhada et al., "Performance Analysis of Long Short-term Memory (LSTM) Model for Remaining Useful Life Prediction on Turbofan Engine," *J. Electron. Technol. Explor.*, vol. 3, no. 1, pp. 24–30, Jun. 2025, doi: 10.52465/joetex.v3i1.585.

10 Conclusion & Future Work

Conclusion

Developed an **uncertainty-aware prognostics framework** for FD001 using quantile LSTM.

Delivered RUL low/median/high predictions with **interval-based confidence ranking.**

Combined performance, uncertainty, and **SHAP explainability** in one decision-facing pipeline.

Demonstrated **dashboard-ready consumption** for operator-style interpretation.

Future Work

Real-time Deployment

Integrate with IoT sensors and streaming pipelines for real-time monitoring and alerts.

Advanced Models

Explore Transformers, attention-based architectures, and hybrid models for improved accuracy and generalization.

Uncertainty Quantification

Improve uncertainty calibration and interval sharpness for more reliable risk assessment.

Scalability

Extend to FD003 (multiple fault modes) and other industrial assets for broader applicability.

Live Demo Dashboard

https://rul-dashboard-app.sarthakchandervanshi.uk/

Interactive FD001 results, uncertainty bands, SHAP insights, and per-engine analysis.

Key Takeaway

This work presents a practical, uncertainty-aware RUL prognostics pipeline that combines accuracy, explainability, and operational relevance–ready for decision-facing applications and extensible to real-world industrial environments.

Note: Results are from a single-dataset prototype (NASA C-MAPSS FD001, last-window-per-test-engine). Further validation on additional datasets and real-world data is future work.

THANK YOU!

Any Questions?

https://github.com/SarthakChandervanshi/Digital-Twin-Framework-for-Industrial-Asset-Prognostics